

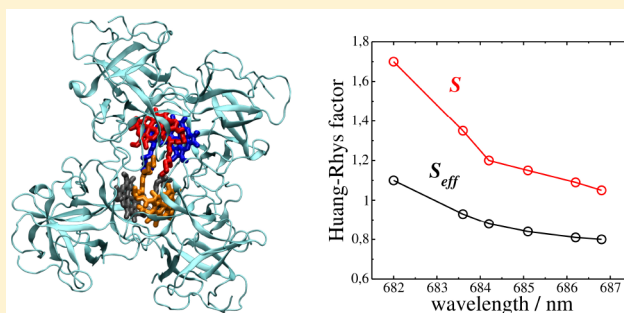
Wavelength-Dependent Exciton–Vibrational Coupling in the Water-Soluble Chlorophyll Binding Protein Revealed by Multilevel Theory of Difference Fluorescence Line-Narrowing

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S Supporting Information

ABSTRACT: One of the most powerful line-narrowing techniques used to unravel the homogeneous lineshapes of inhomogeneously broadened systems is difference fluorescence line-narrowing spectroscopy. When this spectroscopy was applied to multichromophoric systems so far, the spectra were analyzed by an effective two-level system approach, composed of the electronic ground state and the lowest exciton state. An effective Huang–Rhys factor was assigned for the coupling of this state to the vibrations. Here, we extend this approach by including a multilevel line shape theory, which takes into account the excitonic coupling between pigments and thereby the effect of the delocalization of the excited states explicitly. In this way, it becomes possible to extract the spectral density of the local exciton–vibrational coupling. The theory is applied to the recombinant water-soluble chlorophyll binding protein reconstituted with chlorophyll *a* or *b* and reveals a significant decrease of the Huang–Rhys factor of the local exciton–vibrational coupling with decreasing transition energy of the chlorophylls. This decrease could be due to the increase in steric interactions reducing the flexibility of the environment and red-shifting the site energy of the pigments.



INTRODUCTION

The homogeneous line width of pigment–protein complexes is masked by inhomogeneous broadening caused by static disorder in local transition energies of the pigments. In line-narrowing spectroscopy¹ a selection of a subensemble is achieved by narrow-band excitation and the spectrum of this subensemble is analyzed. In fluorescence line-narrowing (FLN) spectroscopy, the fluorescence spectrum of the selectively excited ensemble is detected, exhibiting a sharp zero phonon line (ZPL) and a broad vibrational sideband (so-called phonon sideband, PSB). The latter contains important information about the spectral density of the pigment–protein coupling, which is the central quantity in energy transfer and line shape theories.² From the integral intensities of the PSB relative to that of the ZPL, the Huang–Rhys factor S (in the absence of pigment–pigment coupling) can be determined.³ It is a measure of the overall exciton–vibrational coupling strength. In the presence of pigment–pigment coupling the delocalization of excited states needs to be taken into account in the estimation of S , as demonstrated in this work.

A direct evaluation of the ZPL intensity in FLN spectra is impossible because the fluorescence of the ZPL overlaps with scattered light from the excitation laser. To overcome this problem, one can analyze the dependence of the shape of the PSB on the excitation frequency⁴ or analyze, in parallel to FLN, the temperature dependence of the inhomogeneous absorption spectrum.^{5,6} In both cases it is assumed that neither

S nor $J(\omega)$ varies across the inhomogeneous ensemble. Two methods that do not rely on this assumption and can resolve the intensity of the ZPL, and thereby the excitation wavelength dependence of S and $J(\omega)$, are single molecule spectroscopy (SMS)^{7,8} and difference fluorescence line-narrowing (Δ FLN) spectroscopy.^{3,9–11}

In Δ FLN spectroscopy, FLN is combined with hole-burning (HB) spectroscopy. The FLN spectra are measured before and after exciting the sample with an energetically narrow burn-laser. The latter induces conformational transitions in the proteins, which change the FLN signal. Since the scattered light of the fluorescence excitation is unaffected by the conformational transition, the Δ FLN signal is essentially free of scattered light. So far, Δ FLN spectra were described by using an effective electronic two-level system approach^{3,9–11} representing the ground state and the lowest exciton state of the complex. We have recently presented a multilevel system approach for HB spectra.¹² Here we extend this approach to the description of Δ FLN spectra and use it to uncover the local Huang–Rhys factor of chlorophyll *a* (Chl*a*) and Chl*b* bound to the water-soluble chlorophyll binding protein (WSCP).

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THEORY AND COMPUTATIONAL METHODS

The pigments are described as excitonically coupled two-level systems that interact with intermolecular vibrations. The homogeneous linear absorption and fluorescence spectra $\alpha(\omega)$ and $I(\omega)$, are described by^{4,12–15}

$$\begin{aligned}\alpha(\omega) &\propto \omega \sum_M |\vec{\mu}_M|^2 D_M(\omega) \\ I(\omega) &\propto \omega^3 \sum_M f_M |\vec{\mu}_M|^2 D'_M(\omega)\end{aligned}\quad (1)$$

with the excitonic transition dipole moment $\vec{\mu}_M$ and the Boltzmann factor f_M for the thermal population of exciton states. An expression for the circular dichroism spectrum is given in the [Supporting Information](#), eq S2. The absorption line shape function $D_M(\omega) = D_M^{(0)}(\omega) + D_M^{\text{vib}}(\omega)$ is composed of the symmetric ZPL

$$D_M^{(0)}(\omega) = \frac{1}{\pi} \frac{\tau_M^{-1}}{(\omega - \tilde{\omega}_M)^2 + \tau_M^{-2}} e^{-\gamma_M G(0)} \quad (2)$$

and the asymmetric PSB

$$D_M^{\text{vib}}(\omega) = \frac{1}{2\pi} e^{-\gamma_M G(0)} \int_{-\infty}^{+\infty} dt e^{i(\omega - \tilde{\omega}_M)t} (e^{\gamma_M G(t)} - 1) e^{-|t|/\tau_M} \quad (3)$$

where

$$\gamma_M = \sum_m |c_m^{(M)}|^4 \quad (4)$$

is the inverse participation ratio. It is a measure of the delocalization of exciton state $|M\rangle = \sum_m c_m^{(M)} |m\rangle$, which is given as a linear combination of local excited states $|m\rangle$ of the aggregate. The function $G(t)$ is related to the spectral density $J(\omega)$ and the Bose–Einstein distribution function of vibrational quanta $n(\omega)$ as described in the [Supporting Information](#) eq S7. The τ_M in eqs 2 and 3 is the dephasing time of exciton state $|M\rangle$ and is given as $\tau_M^{-1} = \frac{1}{2} \sum_N^{N \neq M} k_{M \rightarrow N} + \tau_{\text{pd}}^{-1}$ with the Redfield rate constant $k_{M \rightarrow N}$ of exciton relaxation between state $|M\rangle$ and $|N\rangle$, and with τ_{pd}^{-1} describing the pure dephasing. The Redfield rate constant is related to $J(\omega)$ and $n(\omega)$, as explained in the [Supporting Information](#) (eq S4). The line shape function $D'_M(\omega)$ for fluorescence has the same ZPL (eq 2) as the $D_M(\omega)$ of absorption, but the factor $e^{i(\omega - \tilde{\omega}_M)t}$ in eq 3 has to be replaced by $e^{-i(\omega - \tilde{\omega}_M)t}$ ([Supporting Information](#), eq S8).

In HB spectroscopy, the difference between an absorption spectrum after excitation with the burn laser (postburn) and before (preburn) is measured, $\alpha_{\text{HB}}(\omega) = \alpha_{\text{post}}(\omega) - \alpha_{\text{pre}}(\omega)$. In nonphotochemical HB a conformational transition of the protein environment of the pigments occurs when they are in their excited state. The protein stays in this conformational substate after the pigments have come back to their electronic ground state and the linear absorption spectrum of the complex has changed from $\alpha(\omega)$ to $\alpha_m(\omega)$, where the index m indicates the pigment that has changed its site energy due to the conformational transition of its local protein environment. Note that these conformational transitions are so rare that a simultaneous conformational transition of the local environments of two or more pigments can be neglected. The absorption spectrum $\alpha_m(\omega)$ is obtained by assigning a new site energy to pigment m and diagonalizing the new exciton Hamiltonian resulting from this site energy shift. Analogous to

HB, the ΔFLN spectrum is the difference of a FLN spectrum before and after burning. Note the different sign convention used for $\alpha_{\text{HB}}(\omega)$ and $\Delta\Phi(\omega)$. Hence, the expression for the ΔFLN spectrum $\Delta\Phi(\omega)$ is obtained from our earlier expression for the HB signal $\alpha_{\text{HB}}(\omega)$ (eq 14 in ref 12) by switching the sign and replacing the preburn absorption $\alpha(\omega)$ by the selectively excited preburn fluorescence $I(\omega)\alpha(\omega_{\text{exc}})$ and the postburn absorption after resonant and nonresonant burning $\alpha_m^{(\text{res})}(\omega)$ and $\alpha_m(\omega)$ by the selectively excited postburn fluorescence $I_m^{(\text{res})}(\omega)\alpha_m^{(\text{res})}(\omega_{\text{exc}})$ and $I_m(\omega)\alpha_m(\omega_{\text{exc}})$, respectively, resulting in

$$\Delta\Phi(\omega) \propto \Phi_+(\omega) - \Phi_-(\omega) \quad (5)$$

with

$$\Phi_+(\omega) = \langle I(\omega)\alpha(\omega_{\text{exc}})\alpha(\omega_{\text{B}}) \rangle_{\text{dis}} \quad (6)$$

and

$$\begin{aligned}\Phi_-(\omega) &= \omega \left\langle |\vec{\mu}_{01}|^2 \sum_m |c_m^{(1)}|^2 (I_m^{(\text{res})}(\omega)\alpha_m^{(\text{res})}(\omega_{\text{exc}})D_1^{(0)}(\omega_{\text{B}}) \right. \\ &\quad \left. + I_m(\omega)\alpha_m(\omega_{\text{exc}})D_1^{(\text{vib})}(\omega_{\text{B}})) \right\rangle_{\text{dis}} \\ &\quad + \omega \left\langle \sum_m |c_m^{(1)}|^2 I_m(\omega)\alpha_m(\omega_{\text{exc}}) \sum_{M=2}^N |\vec{\mu}_{0M}|^2 D_M(\omega_{\text{B}}) \right\rangle_{\text{dis}}\end{aligned}\quad (7)$$

Here $\langle \dots \rangle_{\text{dis}}$ denotes an average over static disorder in site energies ([Supporting Information](#) eq S10). The complexes that did not change their exciton Hamiltonian during the hole-burning process contribute by $\Phi_+(\omega)$ to the ΔFLN spectrum $\Delta\Phi(\omega)$. In the remaining complexes, which contribute negatively by $\Phi_-(\omega)$ to $\Delta\Phi(\omega)$, the excitation by the burn laser caused a site energy shift of one of the pigments. $I_m(\omega)$ and $\alpha_m(\omega)$ denote the homogeneous fluorescence and absorption spectrum, respectively, of a complex where pigment m has changed its site energy and the superscript “res” indicates that the complex has been excited resonantly by the burn laser, that is, excitation of the zero-phonon transition of the lowest exciton state with line shape $D_1^{(0)}$. In this case the postburn transition frequency ω_m is not taken from the original distribution function of static disorder $P(\omega_m - \bar{\omega}_m, \sigma)$ but instead from a distribution function $P(\omega_m - \omega'_m, \sigma_{\text{res}})$ that is centered around the preburn site energy $\hbar\omega'_m$ of pigment m (memory effect, see [Supporting Information](#) Figure S1). In addition, we assign a smaller width σ_{res} to this distribution function to take into account the reduced conformational flexibility.¹² The second and third terms on the rhs of eq 7 take into account nonresonant burning processes, that is, excitation of the vibrational sideband of the lowest exciton state ($M = 1$) and excitation of the remaining exciton states ($M = 2, \dots, N$) at higher energies. In these cases we found¹² that there is enough excess energy available after excitation, either by exciton relaxation or by vibrational excitation, that all conformational substates of the protein can be reached, and therefore, the preburn distribution function $P(\omega_m - \bar{\omega}_m, \sigma)$ of the transition energy of pigment m applies also to the postburn site energy. Note that, so far, the same excitation and burn frequencies are used in ΔFLN spectroscopy.^{3,9–11} Hence, we have $\omega_{\text{exc}} = \omega_{\text{B}}$ in eqs 6 and 7.

RESULTS AND DISCUSSION

In the following, we apply the present theory to recombinant class IIa WSCP from cauliflower reconstituted with Chla and

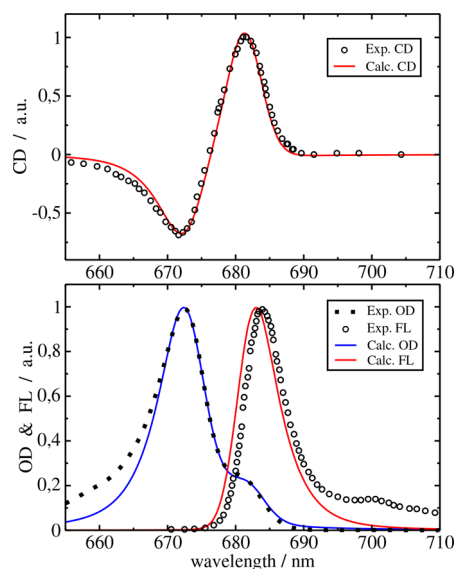


Figure 1. Top: Calculated (red solid line) low temperature (1.7 K) circular dichroism (CD) spectrum of Chla-WSCP compared to experimental data by Hughes et al.³⁰ (black circles). Bottom: Low temperature (4.5 K) linear absorption (OD) and fluorescence (FL) spectra of Chla-WSCP. Black squares and circles represent the experimental absorption and fluorescence spectra, respectively, by Pieper et al.²⁴ The calculated absorption and fluorescence spectra are shown as blue and red solid lines, respectively.

Chlb. Recently, the crystal structure of native class IIa WSCP of cauliflower was obtained with a resolution of 1.9 Å.¹⁶ This structure is highly homologous to the structure of class IIb WSCP complex of *L. virginicum*¹⁷ used before for the calculation of HB spectra,¹² measured on class IIa WSCP.¹⁸ The four Chla pigments identified in the two crystal structures are arranged in two “open sandwich” dimers, where the intradimer couplings are strong and the interdimer couplings are weak. Assuming that the Q_y transition dipoles are oriented along the N_B-N_D axis of the Chls, an opening angle of $\alpha = 30^\circ$ is obtained between the transition dipole moments of the strongly coupled Chls. As shown in recent publications,^{19,20} the fit of inhomogeneous optical spectra of recombinant WSCP reconstituted with Chlb can be improved by increasing the angle α by 6° to 9° . In the present work, we reinvestigated the change in opening angle between transition dipole moments (and thereby the angle between the transition dipole moment and the N_B-N_D axis of the Chls, Supporting Information Figures S2 and S3), resulting in $\alpha = 27^\circ$ and $\alpha = 39^\circ$ for Chla- and Chlb-WSCP, respectively. The calculated inhomogeneous optical spectra, including circular dichroism, linear absorption, and fluorescence of Chla-WSCP at cryogenic temperatures and linear absorption at $T = 4.5, 77$, and 298 K and fluorescence at 4.5 K of Chlb-WSCP, are compared in Figures 1 and 2 with experimental data. The deviations between calculations and experiments in the blue wing of the absorption and in the red wing of the fluorescence spectra are due to high-frequency intramolecular vibrations of the Chls, which were not included in the calculations. Close inspection of the fluorescence spectra reveals a slight blue shift of the

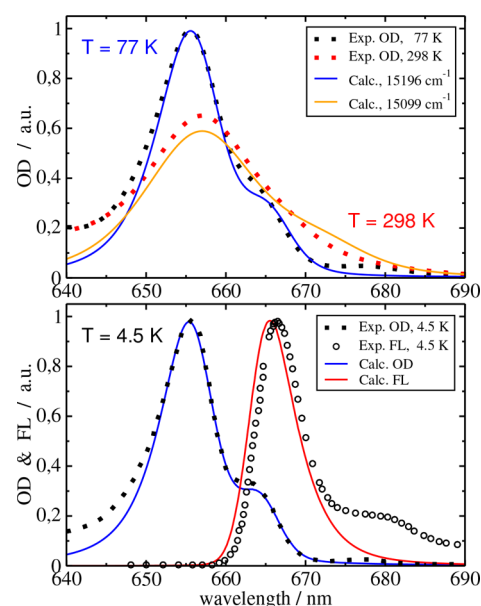


Figure 2. Top: Linear absorption (OD) spectra of Chlb-WSCP at two different temperatures. Black and red squares show the experimental data,²⁵ and blue and orange lines show the calculated spectra at $T = 77$ and 298 K, respectively. In the calculation of the spectra at $T = 77$ and 298 K, a 16 and 113 cm^{-1} red-shift, respectively, of the mean site energy of the Chls with respect to the site energy at $T = 4.5$ K (Table 1) was assumed. Bottom: Low temperature (4.5 K) linear absorption (OD) and fluorescence (FL) spectra of Chlb-WSCP. Black squares and circles represent the experimental absorption and fluorescence spectra, respectively, by Pieper et al.²⁴ The calculated absorption and fluorescence spectra are shown as blue and red solid lines, respectively.

Table 1. Parameters Used for the Calculation of Spectra^a

	Chla-WSCP		Chlb-WSCP	
	ref 12	this work	ref 12	this work
V/cm^{-1}	83	77	82	77
σ/cm^{-1}	64	66	66	66
$\sigma_{\text{res}}/\text{cm}^{-1}$	8	8	8	8
$\hbar\bar{\omega}/\text{cm}^{-1}$	14 808	14 823	15 200	15 212
α/deg	30	27	36	39
S	0.8	1.15	0.8	1.25
$E_{\lambda}/\text{cm}^{-1}$	62.8	95.2	62.8	104.2
$\tau_{\text{pd}}/\text{fs}$ (ΔFLN)		15 000		15 000
$\tau_{\text{pd}}/\text{fs}$ (HB)	2 750	2 750	2 750	2 750

^aThe value of $\sigma = 66$ cm^{-1} corresponds to $\Delta_{\text{inh}} = 2\sqrt{2 \ln 2} \sigma = 155$ cm^{-1} (full width at half maximum). The values of the Huang-Rhys factor S refer to the calculation of inhomogeneous spectra (Figures 1 and 2) and to the ΔFLN spectra in Figure 3. The values of the mean site energy $\hbar\bar{\omega}$ refer to $T < 77$ K.

calculated with respect to the measured spectra. As found in time-resolved fluorescence spectra,²¹ there is a slight red shift of the fluorescence maximum occurring with a 100 ps relaxation time constant, which obviously is not covered by the present theory. A more sophisticated treatment of the system bath entanglement²² and/or a more realistic description of the potential energy surface of the pigment–protein complex^{21,23} might be needed. In principle, this effect could be taken into account implicitly by shifting the electronic transition energy of the pigments in fluorescence. The effect on the ΔFLN spectra, however, is negligible.

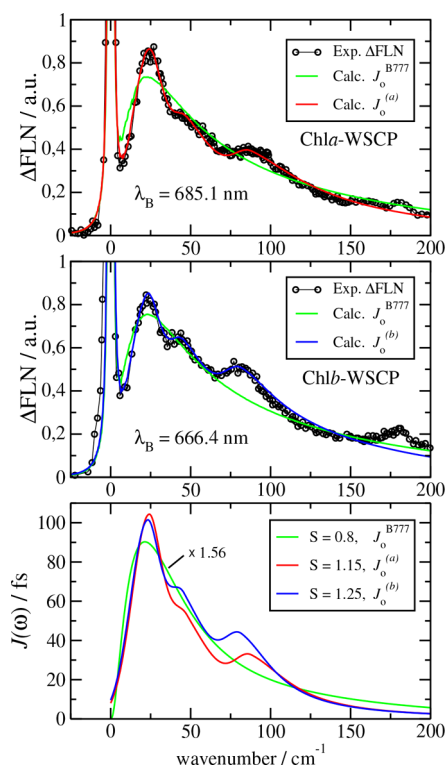


Figure 3. Δ FLN spectra, calculated with the spectral densities $J(\omega)$ shown in the bottom part, are compared with experimental data²⁴ for Chla-WSCP (top) and Chlb-WSCP (center). The green line in the bottom part has been scaled up by a factor of 1.56 for better comparison. The different functional forms of the spectral densities are described in the main text (eqs 9 and 10) and in the Supporting Information (Table S1, eq S11).

The parameters of our model are summarized in Table 1 and comprise, in addition to the angle α between the transition dipole moments of Chls, the excitonic coupling V , the width σ of the inhomogeneous distribution function for the site energies, the width σ_{res} of the distribution function used to assign a postburn site energy for resonant burning, the equal mean site energy $\hbar\bar{\omega}$ of the two Chls in the homodimer, the Huang–Rhys factor S of the local exciton–vibrational coupling of the two Chls, and the pure dephasing time τ_{pd} used to describe the finite width of the ZPL. These parameter values were taken from our earlier work¹² on HB spectra of this system, with slight modifications, which are in large part due to the refinement of the spectral density $J(\omega)$ of the exciton–vibrational coupling from the present Δ FLN spectra. Note that the τ_{pd} for Δ FLN calculations is larger than that used for HB calculations, reflecting broadening mechanisms appearing in HB but not in Δ FLN, probably due to the 2–3 orders of magnitude larger burn fluence in the HB experiments. The spectral density of the local exciton–vibrational coupling $J(\omega)$ is described as

$$J(\omega) = SJ_0(\omega) \quad (8)$$

with the Huang–Rhys factor S and the normalized function $J_0(\omega)$. As possible candidates for the latter, we took the rather structureless spectral density $J_0^{B777}(\omega)$ extracted from FLN experiments of the monomeric B777 complex,⁴ used before in the calculation of HB spectra,¹² and spectral densities $J_0^{(a)}$ and $J_0^{(b)}$ inferred by Pieper et al. for Chla-WSCP and Chlb-WSCP, respectively, from Δ FLN spectra, using an effective two-level

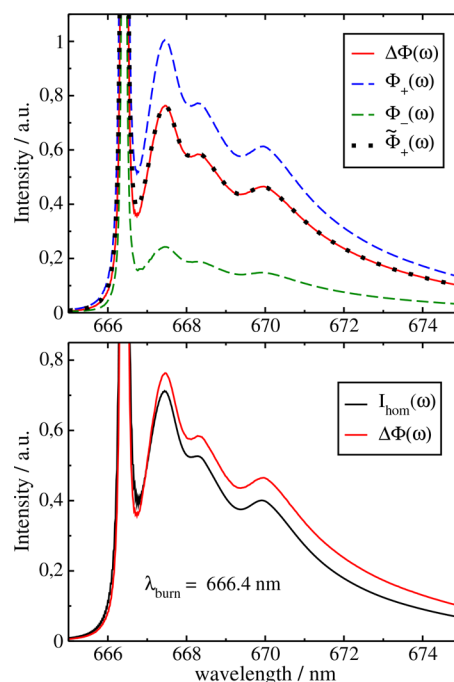


Figure 4. Upper part: The Δ FLN spectrum $\Delta\Phi(\omega) = \Phi_+(\omega) - \Phi_-(\omega)$ (red line) calculated for Chlb-WSCP is compared to its positive $\Phi_+(\omega)$ (eq 6, blue dashed line) and negative $\Phi_-(\omega)$ (eq 7, green dashed line) components, as well as to a function $\tilde{\Phi}_+(\omega) = c\Phi_+(\omega)$, where the scaling constant c was chosen such that the integral over $\tilde{\Phi}_+(\omega)$ equals that over $\Delta\Phi(\omega)$. Lower part: The Δ FLN spectrum of the upper part is compared to the homogeneous fluorescence spectrum $I_{\text{hom}}(\omega)$ calculated for Chlb-WSCP with equal site energies, chosen such that the resulting ZPL is at the same energy as the ZPL of the Δ FLN spectrum.

system approach.²⁴ The functional form of $J_0^{B777}(\omega)$ is given in Supporting Information eq S11, and that of $J_0^{(a)}(\omega)$ and $J_0^{(b)}(\omega)$ reads

$$J_0^{(a/b)}(\omega) = \frac{\sum_{m=1}^3 s_m f_m(\omega)}{\sum_{k=1}^3 s_k} \quad (9)$$

where the low energy part of each term $f_m(\omega)$ is a Gaussian and the high energy part is a Lorentzian function:

$$f_m(\omega) = \begin{cases} \frac{1}{c_m} \exp\left(-\frac{(\omega - \omega_m)^2}{2\sigma_m^2}\right) & \text{if } \omega \leq \omega_m \\ \frac{1}{c_m} \frac{(0.5\Gamma_m)^2}{(\omega - \omega_m)^2 + (0.5\Gamma_m)^2} & \text{if } \omega > \omega_m \end{cases} \quad (10)$$

with different parameter values s_m , σ_m , and Γ_m for Chla-WSCP and Chlb-WSCP, given in Supporting Information Table S1. The functions f_m are normalized such that $\int f_m(\omega) d\omega = 1$ with the normalization factors $c_m = \sqrt{\frac{\pi}{2}} \sigma_m + \frac{\pi}{4} \Gamma_m$. A comparison between the different spectral densities is given in the bottom part of Figure 3. The Δ FLN spectra of Chla-WSCP excited at 685.1 nm and of Chlb-WSCP excited at 666.4 nm, resulting from these spectral densities, are compared to experimental data in the top and middle parts of Figure 3, respectively. Whereas with $J_0^{B777}(\omega)$ only a qualitative description of the Δ FLN spectra is possible, $J_0^{(a)}(\omega)$ and $J_0^{(b)}(\omega)$ describe the spectra quantitatively, if Huang–Rhys factors $S_a = 1.15$ and S_b

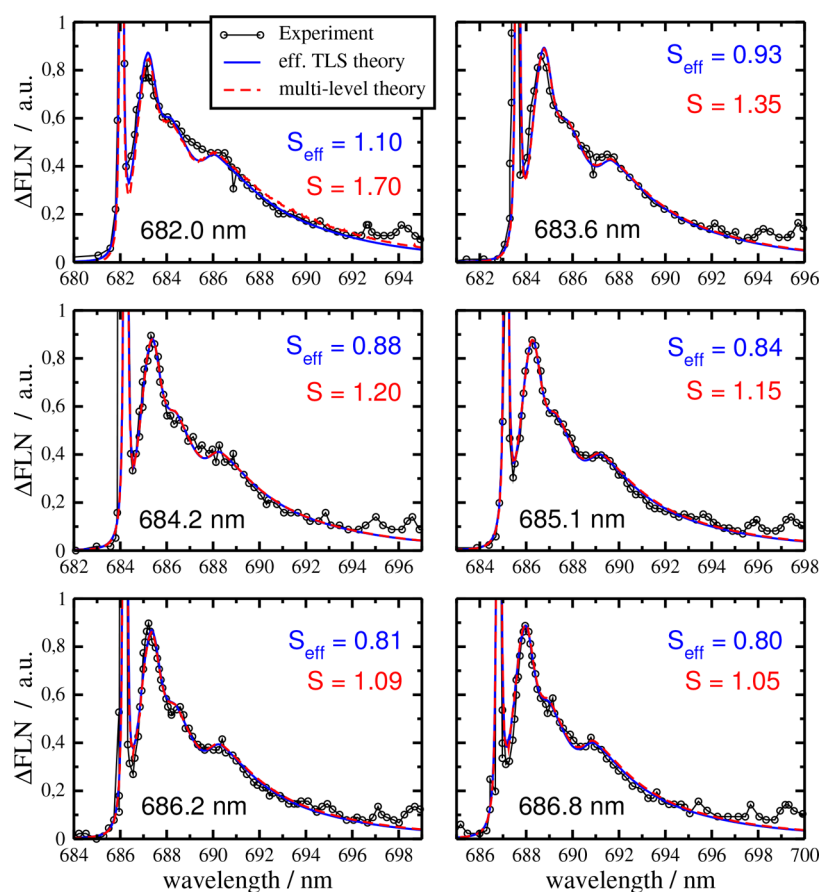


Figure 5. Low temperature (4.5 K) Δ FLN spectra of Chla-WSCP at six different excitation wavelengths given in the figures (black numbers). The experimental data²⁴ (black lines with circles) are compared to calculations using the present multilevel system theory (red dashed lines) with the respective Huang–Rhys factor S (red numbers) and to calculations in the effective two-level system (TLS) approach (blue solid lines) using the respective effective Huang–Rhys factors S_{eff} (blue numbers) from ref 24.

$= 1.25$ are assigned for Chla-WSCP and Chlb-WSCP, respectively.

In the upper part of Figure 4 we compare the Δ FLN spectrum $\Delta\Phi = \Phi_+ - \Phi_-$ of Chlb-WSCP excited at 666.4 nm with its positive and negative contributions Φ_+ and Φ_- , respectively. In addition, a scaled function $\tilde{\Phi}_+ = c\Phi_+$ is shown, where the scaling constant c is chosen such that the integral over Φ_+ equals that over $\Delta\Phi$ ($c = 0.758$). The close agreement between $\tilde{\Phi}_+$ and $\Delta\Phi$ demonstrates that Φ_+ is a very good approximation for the shape of the Δ FLN spectrum. Hence, in striking contrast to hole-burning spectra, where the antihole changes the spectrum qualitatively (Supporting Information Figure S4),¹² here the signal arising from complexes that changed their spectrum is much less important. Therefore, the Δ FLN spectrum is more sensitive to details of the spectral density, which are masked by antihole contributions in HB. Indeed, the HB spectra of WSCP can be described equally well with the present refined spectral density (Supporting Information Figure S4).

The Huang–Rhys factor $S = 0.8$ in connection to $J_0^{B777}(\omega)$ has been assigned previously,²⁵ based on the broadening of the absorption spectrum of Chlb-WSCP between two different temperatures, $T = 77$ and 298 K. The Huang–Rhys factor $S_b = 1.25$ inferred from the Δ FLN spectrum of Chlb-WSCP, Figure 3 (middle part), leads to a somewhat too large thermal broadening of the absorption spectrum at $T = 298$ K, whereas the low-temperature spectra are described well (Figure 2). This

result indicates that the present low-temperature Huang–Rhys factor may be somewhat different at physiological temperatures. Variations of this type are possible, in particular, around the glass transition temperature at roughly 200–250 K, where conformational transitions of the protein occur, as detected, e.g., in the light-harvesting complex LHC-II of higher plants with quasi-elastic neutron scattering and optical spectroscopy.²³ Absorption spectra of WSCP at different intermediate temperatures would be helpful for a more detailed analysis of this effect. Most likely, these conformational transitions are also responsible for the red-shift of the site energy of the Chls in WSCP with increasing temperature (see caption of Figure 2 and ref 25).

The homogeneous fluorescence spectrum, calculated for the mean site energies, is shown together with the Δ FLN spectrum in the lower part of Figure 4. From this comparison we learn that the shape of the vibrational sideband of the Δ FLN spectrum is a good approximation for that of the homogeneous fluorescence, but the relative intensity of the vibrational sideband with respect to that of the ZPL is somewhat different. The relative intensity r of the vibrational sideband in the homogeneous spectrum from eqs 2 and 3 for low temperatures is obtained as

$$r = \frac{\int d\omega D_M^{(\text{vib})}(\omega)}{\int d\omega D_M^{(0)}(\omega)} = e^{\gamma_M S} - 1 \quad (11)$$

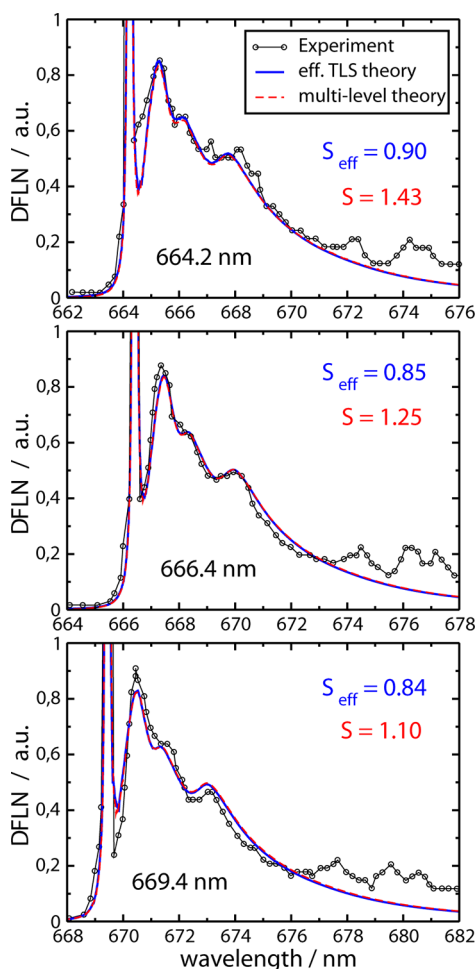


Figure 6. Low temperature (4.5 K) Δ FLN spectra of Chlb-WSCP at three different excitation wavelengths given in the figures (black numbers). The experimental data²⁴ (black lines with circles) are compared to calculations using the present multilevel system theory (red dashed lines) with the respective Huang–Rhys factor S (red numbers) and to calculations in the effective two-level system (TLS) approach (blue solid lines) using the respective effective Huang–Rhys factors S_{eff} (blue numbers) from ref 24.

where γ_M is the inverse participation ratio in eq 4 and we used the fact that for low temperatures; it holds that $G(0) \approx S$. For two coupled pigments, γ_M is between $\frac{1}{2}$ and 1, where $\gamma_M = 1$ means complete localization and $\gamma_M = \frac{1}{2}$ complete delocalization of the excitation. The homogeneous fluorescence spectrum was calculated for a homodimer with equal site energies. Therefore, exciton delocalization is larger and the relative weight of the PSB with respect to the ZPL ($r = e^{S/2} - 1 = 0.9$) is smaller than that seen in the Δ FLN spectrum ($r = 1.3$), where also dimers of Chls with different site energies and, hence less delocalization, contribute. In order to quantify this effect, we introduce the frequency-dependent inverse participation ratio of the lowest exciton state

$$\bar{\gamma}(\omega) = \left\langle \sum_{m=1}^N |c_m^{(1)}|^4 \delta(\omega - \tilde{\omega}_{10}) \right\rangle_{\text{dis}} \quad (12)$$

It provides the average inverse participation ratio of the complexes that have their zero phonon transition of the lowest exciton state at $\omega = \tilde{\omega}_{10}$. For the present excitation wavelength

of 666.4 nm we have $\bar{\gamma} = 0.618$ which results in a relative intensity of the vibrational sideband of $r = e^{\bar{\gamma}S} - 1 = 1.2$, which is close to the $r = 1.3$ of the Δ FLN spectrum.

In the standard analysis of Δ FLN spectra,^{1,3,24} the system is described as an effective electronic two-level system (TLS), representing the ground state and the first exciton state, from where the fluorescence starts. We want to investigate in the following how well such an effective TLS approach works and how the system parameters extracted from such an approach are related to the actual parameters of the exciton Hamiltonian. The energy difference between the two levels obviously has to be identified with the excitation energy of the lowest exciton state, and the dephasing time τ_M in eqs 2 and 3 becomes the pure dephasing τ_{pd} since no exciton relaxation is taken into account. The vibrational sideband $D_1^{\text{vib}}(\omega)$ of the lowest exciton state $|M = 1\rangle$ contains the inverse participation ratio γ_1 of this state (eqs 3 and 4). Since in an effective TLS approach no exciton coefficients are available, γ_1 cannot be calculated but has to be replaced by an effective quantity γ_{eff} that is contained in the effective Huang–Rhys factor $S_{\text{eff}} = \gamma_{\text{eff}}S$, which can be extracted from experimental data. Using this TLS approach, Pieper et al.²⁴ arrived at effective Huang–Rhys factors for various excitation wavelengths in Δ FLN spectra on Chla- and Chlb-WSCP, shown in Figures 5 and 6 (see also Supporting Information Figure S5). We have reanalyzed these spectra with the present multilevel theory. The TLS fits of the experimental data, obtained with the effective Huang–Rhys factors reported by Pieper et al.,²⁴ can be well described by the present multilevel theory revealing the actual Huang–Rhys factors S of the local exciton–vibrational couplings. For Chla-WSCP, S_{eff} decreases from 1.1 at an excitation wavelength $\lambda_{\text{exc}} = 682.0$ nm to 0.8 at $\lambda_{\text{exc}} = 686.8$ nm, whereas the actual S factor decreases from 1.70 to 1.05 in the same excitation interval. Similarly in Chlb-WSCP, S_{eff} decreases from 0.9 at $\lambda_{\text{exc}} = 664.2$ nm to 0.84 at $\lambda_{\text{exc}} = 669.5$ nm and S decreases from 1.43 to 1.10.

Three questions arising from these results are the following: (i) Why is the effective Huang–Rhys factor S_{eff} always smaller than the actual Huang–Rhys factor S of the local exciton–vibrational coupling? (ii) Why does S decrease more strongly than S_{eff} with increasing excitation wavelength? (iii) Why does S decrease with increasing wavelength at all? The answers to (i) and (ii) are given with the help of the excitation frequency-dependent inverse participation ratio of the lowest exciton state, $\bar{\gamma}(\omega_{\text{exc}})$, introduced above (eq 12). As demonstrated in Figure 7, in good approximation S_{eff} can be expressed as $S_{\text{eff}} \approx \bar{\gamma}(\omega_{\text{exc}})S$. Since for N coupled pigments it holds that $\frac{1}{N} < \bar{\gamma}(\omega_{\text{exc}}) < 1$, the effective Huang–Rhys factor is always smaller than the actual S -factor. Since the $\bar{\gamma}(\omega_{\text{exc}})$ in Figure 7 gets larger with increasing excitation wavelength, reflecting the fact that the excited states get more localized in the wings of the distribution function, the difference between S_{eff} and S gets smaller with decreasing excitation wavelength λ_{exc} , explaining the different slopes of S and S_{eff} with respect to λ_{exc} .

In light of the static disorder in site energies, it is not surprising that also the local exciton–vibrational coupling strength, characterized by the Huang–Rhys factor S , depends on the specific realization of disorder. So far, an increase of the Huang–Rhys factor with increasing excitation wavelength was reported from a SMS study⁷ on the LH2 antenna complex of purple bacteria and from Δ FLN spectroscopy on Chla in an organic glass,²⁶ whereas no correlation was seen between the emission wavelength and the Huang–Rhys factor in a SMS

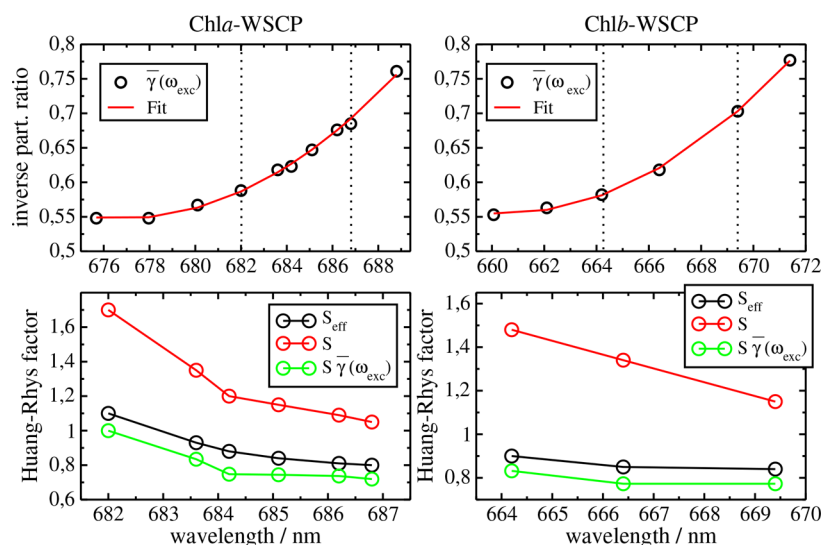


Figure 7. Upper panels: Disorder averaged inverse participation ratio $\bar{\gamma}(\omega_{\text{exc}})$ (eq 12) as a function of excitation wavelength for Chla-WSCP (left) and Chlb-WSCP (right). Lower panels: The Huang–Rhys factor of the local exciton–vibrational coupling S (red) obtained with the present multilevel theory (Figures 5 and 6) as a function of excitation wavelength is compared to the effective Huang–Rhys factor S_{eff} (black) of the lowest exciton state obtained with the effective two-level system approach by Pieper et al.²⁴ and to a function $S\bar{\gamma}(\omega_{\text{exc}})$ (green) with the $\bar{\gamma}(\omega_{\text{exc}})$ from the upper part.

study on photosystem II core complexes.⁸ Hence, to the best of our knowledge, the present system is the first one, for which S decreases with increasing excitation wavelength. What are the microscopic mechanisms that cause the different behaviors of the different systems? At present no quantitative theory exists, but qualitative effects are understood. In LH2 the small interpigment distances give rise to a mixing between charge transfer (CT) and exciton states. Since these CT states are energetically above the exciton states, the mixing causes a red-shift of the latter. The polar nature of the CT states causes a large homogeneous broadening that is partially experienced by the exciton states due to the mixing.⁷ Hence, the stronger is this mixing, the larger is the red shift and the larger is the Huang–Rhys factor of the exciton–vibrational coupling. For Chla in an organic glass most likely the dispersive interaction causes the red-shift of the transition energy and the increase of the homogeneous broadening.²⁶ In the case of photosystem II core complexes electrostatic interactions were found to be responsible for the site energy shifts.^{27,28} Due to the different signs of this interaction (not present, for example, in dispersive interactions), no simple correlation between site energy shift and exciton–vibrational coupling strength is expected. Indeed, normal mode/electrostatic calculations of the spectral density did not show such a correlation.²⁹ For the present WSCP IIa system ring deformations of the Chls caused by the local protein environment were identified as the major factor for the red-shift of the site energy, based on a comparison of the structure and spectra of class IIa and class IIb Chla-WSCP.¹⁶ Class IIa WSCP exhibits a 10 nm red-shift of the absorption spectrum with respect to class IIb WSCP, and the Chls bound in class IIa WSCP have a more ruffled macrocycle. This hypothesis was proven by single point mutations affecting the deformation of the macrocycle of Chls.¹⁶ Along these lines it is likely that the distribution in site energies in the present class IIa WSCP is caused by a distribution in different degrees of ring deformations of the Chls caused by the local protein environments. The larger the steric interactions are with the environment, the larger is the red shift of the site energy. From

the present decrease of the local Huang–Rhys factor with increasing wavelength we have to conclude that, obviously, these steric interactions reduce the flexibility of the local protein environment and thereby the Huang–Rhys factor of the local exciton–vibrational coupling.

CONCLUSIONS

In summary, we have presented a multilevel theory (eqs 5–7) for the analysis of ΔFLN spectra of pigment–protein complexes. The new method explicitly takes into account the excitonic coupling between pigments and allows extraction of the Huang–Rhys factor of the local exciton–vibrational coupling. In contrast to HB spectra analyzed earlier,¹² we find that the contribution $\Phi_{-}(\omega)$ (eq 7) from complexes, in which the burn laser induced a change in the exciton Hamiltonian, to the ΔFLN spectrum $\Delta\Phi(\omega)$ (eq 5) has a negligible influence on the shape of $\Delta\Phi(\omega)$. Hence, in very good approximation it holds that

$$\Delta\Phi(\omega) \propto \langle I(\omega)\alpha(\omega_{\text{exc}})\alpha(\omega_{\text{B}}) \rangle_{\text{dis}} \quad (13)$$

where $I(\omega)$ denotes the fluorescence spectrum and $\alpha(\omega_{\text{exc}})$ and $\alpha(\omega_{\text{B}})$ are the absorption spectra of the complex taken at the excitation and burn frequencies ω_{exc} and ω_{B} , respectively, and $\langle \dots \rangle_{\text{dis}}$ denotes an average over static disorder in site energies. We imagine that the above expression will be helpful also in future analyses of ΔFLN spectra with more sophisticated multilevel line shape theories.

Application of the present method to WSCP reveals a significant decrease of the local Huang–Rhys factor with decreasing transition energy of the pigments. This decrease was masked in part in the earlier TLS analysis²⁴ by the increase of exciton localization with decreasing wavelength, treated implicitly. A possible explanation for the decrease of S with increasing red-shift of the Chl transition energies could be the increase in steric interactions causing a deformation of the Chl macrocycle that was shown to cause the red-shift.¹⁶ According to our hypothesis, these steric interactions cause a decrease in the flexibility of the local protein environment of the Chls, and

in this way the dynamic modulation of transition energies gets less. In order to check this idea, Δ FLN experiments on class IIb WSCP complexes would be helpful. The Chls in these complexes experience smaller ring deformations¹⁶ and hence should show, if at all, a weaker correlation between the site energy shift and the Huang–Rhys factor.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.8b08410.

Theoretical details, additional figures and data (PDF)

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Notes

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